

NB27 Optimization Report

Weighted F1 + Threshold + Missing Robustness + Label Noise

Generated: 2026-06-21 15:33

Executive Summary

NB27 implements 4 optimization GÖREVs from to-do.md:

- 1. Ağırlıklı F1 (weighted): Deterministic evaluation on 80/20 distribution
- 2. Threshold correction: Compare raw F1-max, weighted F1-max, bootstrap robust
- 3. Label noise: Cross-model consensus (3 models) + cleanlab (if available)
- 4. Missing robustness: 4 strategies (M3, no-flags, KNN, sentinel)

PAH baseline: NB21 P4_COMBINED_BalBag Boot F1=0.582

CFTR baseline: NB20 S0c_COMBINED Boot F1=0.863

GÖREV 1+2: Threshold Methods Comparison

Three threshold selection strategies evaluated on PAH and CFTR:

PAH Results

Method	Thr	F1 Boot	±Std	MCC
raw_f1_max	0.160	0.560	0.044	0.497
weighted_f1_max	0.880	0.233	0.127	0.140
boot_8020_robust	0.420	0.590	0.073	0.401

CFTR Results (WARNING: n=21 benign → wide CI)

Method	Thr	F1 Boot	±Std	MCC
raw_f1_max	0.240	0.723	0.110	0.586
weighted_f1_max	0.760	0.546	0.220	0.314
boot_8020_robust	0.230	0.723	0.110	0.586

GÖREV 4: Missing Value Robustness

4 imputation strategies tested: M3 (baseline), no-flags, KNN, sentinel

PAH Missing Robustness

Strategy	F1 Boot	MCC	Prec	Rec
m3	0.233	0.140	0.962	0.163
no_flags	0.233	0.140	0.962	0.163
knn	0.156	0.118	0.971	0.107
sentinel	0.457	0.252	0.967	0.381

CFTR Missing Robustness

Strategy	F1 Boot	MCC	Prec	Rec
m3	0.546	0.314	1.000	0.367
no_flags	0.546	0.314	1.000	0.367
knn	0.391	0.254	1.000	0.267
sentinel	0.258	0.184	1.000	0.156

GÖREV 3: Label Noise Detection (PAH)

Cross-model consensus (3 models: LightGBM, XGBoost, CatBoost)

Suspicious samples (all 3 models wrong): 49/369

Label distribution of suspicious samples:

Pathogenic: 36, Benign: 13

Visualizations

Figure 1: PAH Weighted F1 vs Threshold Sweep

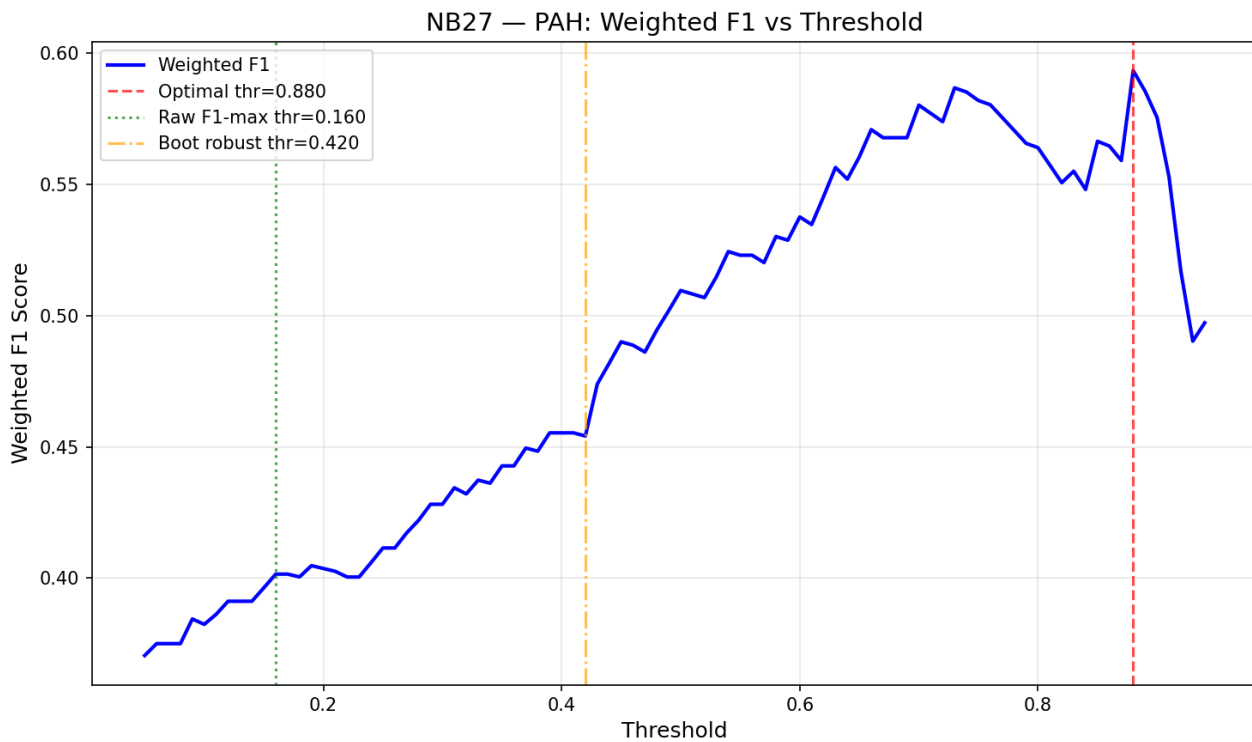


Figure 2: PAH Missing Robustness

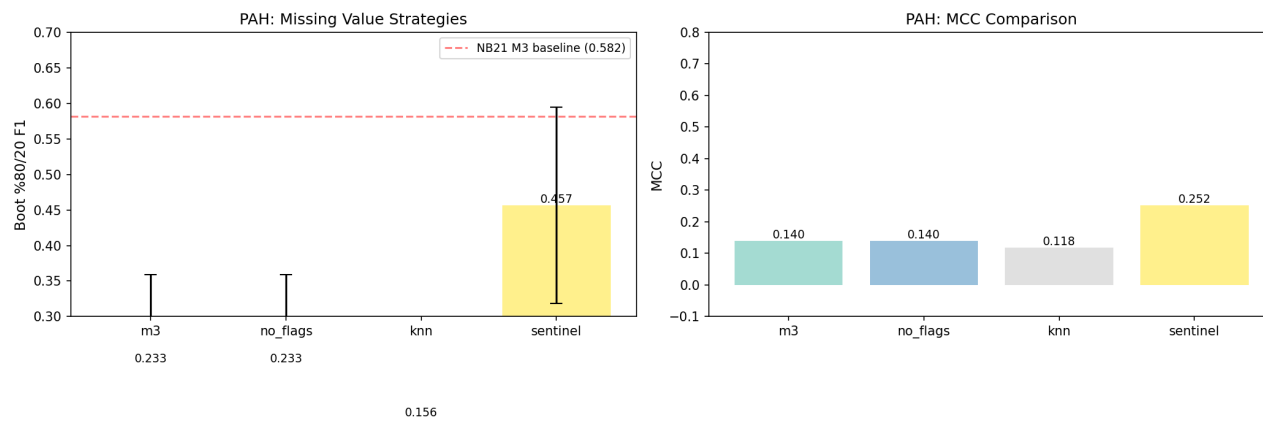


Figure 3: CFTR Missing Robustness

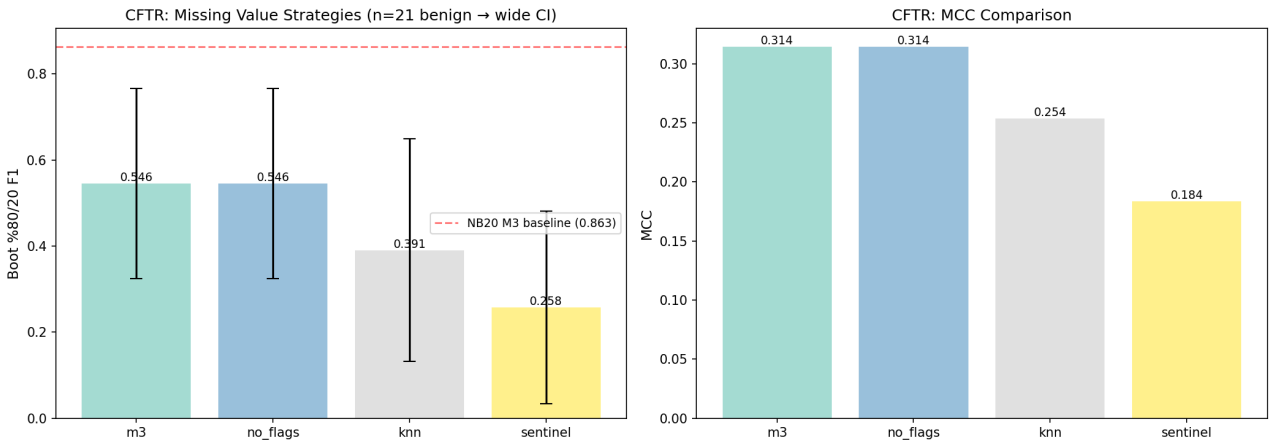


Figure 4: PAH Label Noise Cross-Model Heatmap

